RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
# INDUSTRIAL ENGINEERING & MANAGEMENT

| | | |
|---|---|---|
| **Date : 17/12/2025** | **CIE – II** | **Max. Marks : 10 + 50** |
| **Semester : V** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: Enterprise Information Systems** | | **Course Code : IM355TBB** |

Note:
1. Answer all questions

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1. | Define Robotic Process Automation (RPA) and state its primary focus. | 2 | 1 | CO1 |
| 2. | State the fundamental reason why a company must engage in Business Process Reengineering (BPR) before implementing any major automation initiative. | 2 | 2 | CO2 |
| 3. | Define the term Data Mart and state how it differs in **scope** from a full Data Warehouse. | 2 | 1 | CO3 |
| 4. | Name the two foundational processes that are typically executed by Data Mining tools to discover new knowledge within a BI system. | 2 | 2 | CO3 |
| 5. | A decision-maker uses a BI tool to analyze sales by clicking on a region, then a city, and finally a specific store. Name this OLAP technique and briefly state its purpose. | 2 | 2 | CO3 |
| | **Part – B** | | | |
| 1 | List and explain various methods of business process automation. | 10 | 2 | CO2 |
| 2 | Define Robotic Process Automation (RPA). Explain how it works. List the uses and benefits. | 10 | 2 | CO2 |
| 3 | Explain the core difference between Business Intelligence (BI) and Data Warehousing. Describe why BI is considered “the technology and practice of applying information to make decisions”. | 10 | 2 | CO1 |
| 4 | Describe the four progressive types of Data Analytics used in an EIS. For each type, state the central question it is designed to answer. | 10 | 3 | CO3 |
| 5 | Apply the concept of Online Analytical Processing (OLAP) by explaining its primary function. Illustrate how a retail manager would use an OLAP technique like Drilling Down to analyze a regional sales report. | 10 | 4 | CO3 |

## Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
## INDUSTRIAL ENGINEERING & MANAGEMENT

| | | | |
|---|---|---|---|
| **Date** | **9th January 2025** | **Maximum Marks** | **50** |
| **Course Code** | **IM355TBD** | **Duration** | **90 Min** |
| **Sem** | **V** | **CIE – 2** | |
| **ADVANCED DECISION MODELLING (Professional Core Elective)** | | | |

Note:
1. Answer all the Questions.

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part A** | | | |
| 1. | Mention the situations that make the replacement of items necessary? | 02 | 2 | 1 |
| 2. | What are the three strategies of replacement of items which follow sudden failure mechanisms? | 02 | 2 | 3 |
| 3. | State two differences between preventive maintenance and breakdown maintenance. | 02 | 2 | 2 |
| 4. | Give one advantage and one disadvantage of individual replacement policies. | 02 | 2 | 2 |
| 5. | State the main objectives of using the EOQ model? | 02 | 1 | 3 |
| | **Part B** | | | |
| 1 | A firm is considering the replacement of a machine, whose cost price is Rs 12,200, and its scrap value is Rs 200. From experience the running (maintenance and operating) costs are found to be as follows:<br>Year: 1 2 3 4 5 6 7 8<br>Running Cost (Rs): 200 500 800 1200 1800 2500 3200 4000<br>When should the machine be replaced? | 10 | 3 | 1 |
| 2 | The data on the running costs per year and resale price of equipment A, whose purchase price is Rs 2,00,000 are as follows:<br>Year: 1 2 3 4 5 6 7<br>Running cost (Rs): 30,000 38,000 46,000 58,000 72,000 90,000 1,10,000<br>Resale value (Rs): 1,00,000 50,000 25,000 12,000 8,000 8,000 8,000<br>i. What is the optimum period of replacement?<br>ii. When equipment A is two years old, equipment B, which is a new model for the same usage, is available. The optimum period for replacement is 4 years with an average cost of Rs 72,000. Should equipment A be changed with equipment B? If so, when? | 10 | 3 | 2 |

## Academic year 2024-2025 (ODD Sem)

<table>
<tr><td>3</td><td>A company is considering the purchase of a new machine at Rs 15,000. The economic life of the machine is expected to be 8 years. The salvage value of the machine at the end of the life will be Rs 3,000. The annual running cost is estimated to be Rs 7,000.<br>i. Assuming an interest rate of 5 per cent, determine the present worth of future costs of the proposed machine.<br>ii. Compare the new machine with the presently-owned machine that has an annual operating cost of Rs 5,000 and cost of repair Rs 1,500 in the second year, with an annual increase of Rs 500 in the subsequent years of its life.</td><td>10</td><td>3</td><td>3</td></tr>
<tr><td>4</td><td>The following failure rates have been observed for a certain type of bulbs.
<table>
<tr><td>End of Week:</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr>
<tr><td>Probability of failure to date:</td><td>0.05</td><td>0.13</td><td>0.25</td><td>0.43</td><td>0.68</td><td>0.88</td><td>0.96</td><td>1.0</td></tr>
</table>
The cost of replacing an individual bulb is Rs. 2.25. if the cost of group replacement is 60 paisa per bulb and the total number of bulbs is 1000. What is best interval between group replacements?</td><td>10</td><td>3</td><td>3</td></tr>
<tr><td>5</td><td>Differentiate between Continuous Review and Periodic Review Inventory Systems by the following aspects:<br>i. Monitoring<br>ii. Reorder Trigger<br>iii. Stockouts Risks<br>iv. System Complexity<br>v. Suitability</td><td>10</td><td>2</td><td>2</td></tr>
</table>

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

| **Particulars** | | | CO1 | CO2 | CO3 | CO4 | L1 | L2 | L3 | L4 | L5 | L6 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Marks Distribution** | Quiz | Max Marks | 1 | 2 | 2 | – | 1 | 2 | 2 | – | – | – |
| | Test | Max Marks | 10 | 20 | 20 | – | – | 10 | 40 | – | – | – |

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# RV COLLEGE OF ENGINEERING®, Bengaluru-59

**(Autonomous Institution Affiliated to VTU, Belagavi)**

| | | | | |
|---|---|---|---|---|
| **Program** | CHEMICAL ENGINEERING | | | |
| **Course Title** | GREEN AND HYDROGEN TECHNOLOGY | | | |
| **Course Code** | CH375TGD | | | |
| CONTINUOUS INTERNAL EVALUATION - 2 | | | | |
| GLOBAL ELECTIVE: GROUP D | | | | |
| Sem | V | **Date \| Time** | **DEC 2025** | |
| Max Marks | **60** | **16.12.2025 \| 13:20 - 15:20** | **Max Duration** | **120 Min** |

**Instructions to Candidates:**

All the questions are compulsory. No additional booklet will be provided.

| Q. No. | QUESTIONS | M | BT | CO |
|---|---|---|---|---|
| 1.1 | What are the outcomes of hydrogen risk assessment? | 02 | 01 | 01 |
| 1.2 | List the steps involved in the quantitative hydrogen risk assessment. | 02 | 01 | 04 |
| 1.3 | Define Codes of Practice. | 02 | 01 | 03 |
| 1.4 | What is PAS? | 02 | 01 | 02 |
| 1.5 | List the parameters associated with hydrogen for which the codes and standards are defined. Hint: The approval for these parameters vary between countries. | 02 | 01 | 02 |
| Part B | | | | |
| 2 | Explain in detail, how hydrogen gas is stored as compressed gas including its advantages and disadvantages. | 10 | 02 | 04 |
| 3 | What are the salient features of storing hydrogen as cryogenic liquid? How is it done and what are the associated benefits and hurdles? | 10 | 02 | 04 |
| 4 | Is storing hydrogen as a metal hydride a boon or a bane (annoyance)? Justify with substantial details including the working principle, advantages and disadvantages. | 10 | 03 | 03 |
| 5 | How are liquid organic hydrogen carriers beneficial in case of storage and transport of hydrogen? Provide a detailed narration. | 10 | 02 | 03 |
| 6 | Give an explanatory note on the codes and standards (national and international) associated with the handling and safety of hydrogen gas. | 10 | 03 | 01 |

Course Outcomes

| At the end of the course the student will be able to, | |
|---|---|
| 1 | Understand the importance of hydrogen and its use as an energy carrier |
| 2 | Explain the production, storage and handling of hydrogen |
| 3 | Analyze the need for hydrogen as an alternate fuel and the associated challenges |
| 4 | Appraise the importance of safety, regulations and codes |

| Cos/BTL | CO1 | CO2 | CO3 | CO4 | L1 | L2 | L3 | L4 | L5 | L6 |
|---|---|---|---|---|---|---|---|---|---|---|
| Marks | 12 | 04 | 22 | 22 | 10 | 30 | 20 | . | . | . |

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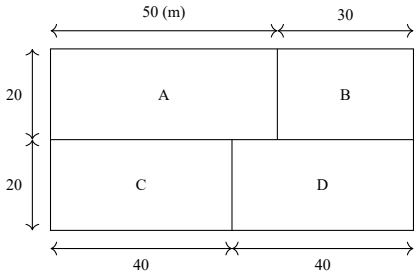
DEPARTMENT OF
# INDUSTRIAL ENGINEERING & MANAGEMENT

| | | |
|---|---|---|
| **Date : 25th May 2026** | **CIE – II** | **Max. Marks : 10 + 50** |
| **Semester : VI** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: Facilities Planning And Design (Professional Core Elective)** | | **Course Code : IM365TDA** |

Note:
1. Answer all the Questions.

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1. | State the main objectives of material handling? | (02) | 1 | 1 |
| 2. | What is meant by unit load design? | (02) | 2 | 2 |
| 3. | Enumerate the importance of flexibility in material handling systems. | (02) | 3 | 3 |
| 4. | What is the role of conveyors in material handling? | (02) | 2 | 4 |
| 5. | State two objectives of computer aided layout techniques. | (02) | 2 | 3 |
| | **Part – B** | | | |
| 1. | Discuss the scope of material handling in modern industries. | (10) | 2 | 2 |
| 2. | Explain movement, time, quantity and space of a material handling system | (10) | 3 | 2 |
| 3. | Explain the following cost components used in estimating material handling cost:<br>i. Equipment cost<br>ii. Storage & Inventory Costs<br>iii. Damage/Wastage Costs | (10) | 3 | 3 |
| 4. | Explain the safety considerations to be followed in material handling operations. | (10) | 3 | 4 |

| | | | | |
|---|---|---|---|---|
| 5. | Consider the following data with unit cost matrix. Use CRAFT pair-wise interchange technique to obtain the desirable layout.<br>Total number of departments: 4<br>Total number of interchangeable departments: 4<br>Initial Layout: | (10) | 2 | 4 |




Area of Departments (Scale 1000 sq units = 1 sq unit):

| Department | A | B | C | D |
|---|---|---|---|---|
| Area (sq. units) | 10 | 6 | 8 | 8 |

Flow Matrix:

| From\To | A | B | C | D |
|---|---|---|---|---|
| A | – | 2 | 4 | 4 |
| B | 1 | – | 1 | 3 |
| C | 2 | 1 | – | 2 |
| D | 4 | 1 | 0 | – |

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DEPARTMENT OF
# INDUSTRIAL ENGINEERING & MANAGEMENT

| | | |
|---|---|---|
| **Date : 25-05-2026** | **CIE – II** | **Max. Marks : 10 + 50** |
| **Semester : VI** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: Service Operations Management** | | **Course Code : IM365TDB** |

Note:
1. Answer all the Questions.

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1. | Imagine you are going to watch an IPL cricket match at Namma Bengaluru's very own Chinnaswamy Cricket stadium, list 4 aspects that you associate regarding service experience. | 2 | 1 | 1 |
| 2. | Assume you get in to an "IT enabled service" (ITES) industry in the next five years. Comment on how you intend to use Service as a competitive advantage. | 2 | 4 | 3 |
| 3. | Assume you are being interviewed for your short but successful career in the field of "Service operations management". One of the questions from a young enthusiast is "How to develop a culture of service quality in an organization?" Comment any two aspects. | 2 | 4 | 2 |
| 4. | Some of you have already started preparing for campus placements. In a year from now, many of your juniors would want your suggestions **(If placed)** and help in many matters of campus placements. To cater to this need and based on some of the requirement analysis that you have done, you wish to start a service and name it as "**Placement war room**" for engineering graduate students. Now answer the following questions.<br>1. Will it be online or offline and why?<br>2. What are the unmet needs of graduate students that your service will fulfil? | 2 | 4 | 4 |
| 5. | If your "**Placement war room**" idea clicks and is appreciated by your initial target audience. Identify any 2-success metrics of your service? Also, comment on the possible perception expectation gap for your service. | 2 | 4 | 3 |
| | **Part – B** | | | |
| 1 | Discuss the role of technology in developing the customer Experience. Cite two examples of services where technology has been used effectively to enhance customer experience. | 10 | 3 | 4 |
| 2 | The nature of supply chains can be very messy and complex. Describe the key issues service operations managers should focus on with a key focus on managing through intermediaries. | 10 | 3 | 4 |
| 3 | Write short notes on<br>a) Service level agreements (SLA)<br>b) Difference between SLA and Operational Level Agreements (OLA) | 5*2=10 | 2 | 3 |
| 4 | Envisage/visualize an experience map for a customer visiting a typical dental clinic. | 10 | 4 | 2 |

| | | | | |
|---|---|---|---|---|
| 5 | One of the key decisions for procurement function is choosing/selecting a supply partner and particularly for those who handle critical elements of supply chain. Discuss the criteria for supplier selection excluding Cost and Quality. | 10 | 3 | 4 |

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| | | |
|---|---|---|
| **Date : 25/05/2026** | **CIE – II** | **Max. Marks : 10 + 50** |
| **Semester : VI** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: Design of Experiments** | | **Course Code : IM365TDD** |

Instructions to Students:
1. Answer all questions
2. Use of statistical tables permitted

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1. | Define "Sparsity of Effects Principle." | 2 | 1 | CO1 |
| 2. | If you have a $2^5$ design, how many total treatment combinations are required for a single replicate? | 1 | 1 | CO1 |
| 3. | Compare the $2^2$ design and $3^2$ designs in terms of modeling. | 2 | 1 | CO2 |
| 4. | What is "Confounding" in factorial experiments? | 2 | 2 | CO2 |
| 5. | Define "Aliasing" in fractional factorial designs. | 1 | 2 | CO2 |
| 6. | Evaluate the risk of using a Resolution III design for a process where two-factor interactions are known to be significant. | 2 | 2 | CO2 |
| | **Part – B** | | | |
| 1 | An engineer is interested in the effects of cutting speed ($A$), tool geometry ($B$), and cutting angle ($C$) on the life (in hours) of a machine tool. Two levels of each factor are chosen, and three replicates of a $2^3$ factorial design are run. The results are as follows: | 20 | 2 | CO4 |

| $A$ | $B$ | $C$ | Treatment Combination | Replicates | | |
|---|---|---|---|---|---|---|
| | | | | I | II | III |
| − | − | − | (1) | 22 | 31 | 25 |
| + | − | − | a | 32 | 43 | 29 |
| − | + | − | b | 35 | 34 | 50 |
| + | + | − | ab | 55 | 47 | 46 |
| − | − | + | c | 44 | 45 | 38 |
| + | − | + | ac | 40 | 37 | 36 |
| − | + | + | bc | 60 | 50 | 54 |
| + | + | + | abc | 39 | 41 | 47 |

i. Estimate the factor effects. Which effects appear to be large?
ii. Use the analysis of variance to confirm your conclusions for part (i).

<table>
<tr><td>2</td><td>The effects of developer strength ($A$) and development time ($B$) on the density of photographic plate film are being studied. Three strengths and three times are used, and four replicates of a $3^2$ factorial experiment are run. The data from this experiment follow. Analyze the data using the standard methods for factorial experiments.
<table>
<tr><td colspan="2" rowspan="2"></td><td colspan="12">Development Time (minutes)</td></tr>
<tr><td colspan="4">10</td><td colspan="4">14</td><td colspan="4">18</td></tr>
<tr><td rowspan="3">Developer Strength</td><td>1</td><td>0</td><td>2</td><td>1</td><td>3</td><td>2</td><td>5</td><td>4</td><td>4</td><td>2</td><td>3</td><td>2</td><td>5</td></tr>
<tr><td>2</td><td>4</td><td>6</td><td>6</td><td>8</td><td>9</td><td>10</td><td>5</td><td>4</td><td>2</td><td>4</td><td>6</td><td>7</td></tr>
<tr><td>3</td><td>7</td><td>7</td><td>5</td><td>7</td><td>7</td><td>8</td><td>5</td><td>7</td><td>10</td><td>8</td><td>9</td><td>12</td></tr>
</table></td><td>20</td><td>3</td><td>CO4</td></tr>
<tr><td>3</td><td>List and explain the three key ideas / principles that are to be implemented for the successful use of fractional factorial designs.</td><td>10</td><td>2</td><td>CO2</td></tr>
</table>

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DEPARTMENT OF
# INDUSTRIAL ENGINEERING & MANAGEMENT

| | | |
|---|---|---|
| **Date : 6th June 2025** | **CIE – 2** | **Max. Marks : 10 + 50** |
| **Semester : VI** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: FACILITIES PLANNING AND DESIGN** | | **Course Code : IM365TDA** |

Note: Answer all the Questions.

**PART A**

| Q No.1 | Questions | M | BT | CO |
|---|---|---|---|---|
| 1.1 | Given station times:<br>• Station 1: 5 min<br>• Station 2: 4 min<br>• Station 3: 6 min<br>Find the bottleneck. | 01 | 2 | 1 |
| 1.2 | Total task time = 20 minutes, Number of workstations = 4, Cycle time = 6 minutes<br>Find line efficiency. | 02 | 1 | 1 |
| 1.3 | Calculate Throughput Time. Given:<br>• Station A: 2 min<br>• Station B: 3 min<br>• Station C: 4 min<br>What is the total throughput time for one product? | 02 | 2 | 1 |
| 1.4 | If a production unit has limited floor space and needs to allow a single operator to handle multiple machines easily, which layout (S-flow or U-flow) is better? | 01 | 1 | 1 |
| 1.5 | Define a unit load as per Tanchoco et al. | 02 | 2 | 1 |
| 1.6 | List any two basic methods of moving the unit load according to Apple. | 01 | 1 | 1 |

**Part –B**

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| 1 | State and explain the different types of material flow with suitable examples. | 10 | 2 | 1 |

<table>
<tr><td>2</td><td>A factory is designing a new assembly line for producing electric scooters. The production involves 5 sequential tasks with the following task times (in minutes):<br><table><tr><th>Task</th><th>Time (min)</th></tr><tr><td>A</td><td>4</td></tr><tr><td>B</td><td>6</td></tr><tr><td>C</td><td>5</td></tr><tr><td>D</td><td>3</td></tr><tr><td>E</td><td>2</td></tr></table>The management is considering either a Straight Line Flow layout or a U-Flow layout. Tasks:<br>a) Calculate the total time required to complete one unit.<br>b) Identify the bottleneck in the process.<br>c) If the total cycle time is 6 minutes and there are 3 workstations, compute the line efficiency.<br>d) Which layout (S-Flow or U-Flow) would you recommend if space is limited and one operator needs to handle multiple stations? Justify your answer</td><td>10</td><td>3</td><td>3</td></tr>
<tr><td>3</td><td>What are the Factors for Consideration in Planning Material Flow</td><td>10</td><td>2</td><td>2</td></tr>
<tr><td>4</td><td>Enumerate and elaborate the steps in material handling system design.</td><td>10</td><td>1</td><td>2</td></tr>
<tr><td>5</td><td>Explain the guidelines for estimating the material handling costs in practice.</td><td>10</td><td>1</td><td>1</td></tr>
</table>

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# INDUSTRIAL ENGINEERING & MANAGEMENT

| | | |
|---|---|---|
| **Date : 6-6-2025** | **CIE – 2** | **Max. Marks : 10 + 50** |
| **Semester : VI** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: Service Operations Management** | | **Course Code : IM365TDB** |

Note:
1. Answer all the Questions.

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1. | Think of a time when you recently received poor service. If service operations are so important, identify reasons why poor service is delivered sometimes? | 2 | 2 | 1 |
| 2. | Assume you get in to an "IT enabled service" (ITES) industry in the next five years. Comment on how you intend to use Service as a competitive advantage. | 2 | 4 | 3 |
| 3. | What is the service concept of "Cyber nurseries". | 2 | 2 | 2 |
| 4. | Comment on the importance of service process design. | 2 | 3 | 4 |
| 5. | From your experience, give two examples where technology is used to enhance customer experience. | 2 | 2 | 2 |
| | **Part – B** | | | |
| 1 | Discuss the role of technology in developing the customer Experience. Cite two examples of services where technology has been used effectively to enhance customer experience. | 10 | 3 | 4 |
| 2 | The nature of supply chains can be very messy and complex. Describe the key issues service operations managers should focus on with a key focus on managing through intermediaries. | 10 | 3 | 4 |
| 3 | Write short notes on<br>a) Service level agreements (SLA)<br>b) Difference between SLA and Operational Level Agreements (OLA) | 10 | 2 | 3 |
| 4 | Visualize an experience map for a customer visiting a typical dental clinic. | 10 | 4 | 2 |
| 5 | One of the key decisions for procurement function is choosing/selecting a supply partner and particularly for those who handle critical elements of supply chain. Discuss the criteria for supplier selection excluding Cost and Quality. | 10 | 4 | 4 |